

Choose large t such that

$$\underline{x} = \underline{x}^* + \frac{z_2 D^{-1} x^{i,2}}{z_2^T D^{-1} z_2}$$

The rate constant matrix can be obtained as

$$\underline{K} = \underline{Z} \underline{\Lambda} \underline{Z}^{-1} \quad (131)$$

Where

$$\underline{Z} = \begin{bmatrix} z_1 & z_2 & \dots & z_n \end{bmatrix}$$

$$\underline{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

*Parameter estimation in non-linear kinetics

Consider a general non-linear (non-elementary) reaction

$$\sum_{j=1}^n \alpha_j A_j = 0$$

Mole balance for constant volume reaction

$$\frac{dC_j}{dt} = \alpha_j r, \quad j = 1, 2, \dots, n$$

But $C_j = C_{j,0} + \alpha_j \xi$

$$\rightarrow \alpha_j \frac{d\xi}{dt} = \alpha_j r \rightarrow \frac{d\xi}{dt} = r$$

In general, for one reaction

$$\frac{d\xi}{dt} = r(\xi, k_1, k_2, \dots, k_p)$$

P parameters of rate expression

If we know $k_1, k_2, \dots, k_p$, then we can calculate $\xi(t)$

Now suppose we have measured values of molar extent per volume.

$\hat{\xi}_1, \hat{\xi}_2, \dots, \hat{\xi}_N$ at different time $t_1, t_2, \dots, t_N$

And objective is to estimate $k_1, k_2, \dots, k_p$ so that measured $\hat{\xi}_j$ are close to fitted ξ_j

$$\text{Min } (k_1, k_2, \dots, k_p) \sum_{i=1}^N [\xi(t_i, k_1, k_2, \dots, k_p) - \hat{\xi}_i]^2 \quad (133)$$

This is a nonlinear least square approximation

Richard E. Bellman (father of dynamic programming) applied the method of quasilinearization to convert (133) to a linear least square problem.

→ Define a vector $\underline{x}$ of n-dimension

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \xi \\ k_1 \\ \vdots \\ k_p \end{bmatrix} \quad (134)$$

Further let

$$\underline{f} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} = \begin{bmatrix} r(\xi, k) \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (135)$$

We can write a set of differential equations

$$\frac{d\underline{x}}{dt} = \underline{f}(\underline{x}) \quad (136)$$

We want to find initial vector $\underline{x}(0)$

$$\underline{x}(0) = \begin{bmatrix} \xi(0) \\ k_1 \\ \vdots \\ k_p \end{bmatrix}$$

→ start with an initial guess

$$\underline{x}(1)(t) = \begin{bmatrix} 0 \\ 1 \times 10^{-5} \\ \vdots \\ 1 \times 10^{-5} \end{bmatrix} \text{ arbitrary}$$

The next iteration $\underline{x}^{(2)}$ can be obtained by solving

$$\frac{d\underline{x}^{(2)}}{dt} = \underline{f}(\underline{x}^{(2)})$$

Expand $\underline{f}(\underline{x}^{(2)})$ about $\underline{x}^{(1)}$ [Taylor series]

$$\frac{d\underline{x}^{(2)}}{dt} = \underline{f}(\underline{x}^{(1)}) + \underline{J} \underline{f}(\underline{x}^{(1)}) [\underline{x}^{(2)} - \underline{x}^{(1)}]$$

(neglect higher order term)

$$\underline{J} = \underline{J} \underline{f} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial r}{\partial \xi} & \frac{\partial r}{\partial k_1} & \dots & \frac{\partial r}{\partial k_p} \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

$$\frac{d\underline{x}^{(2)}}{dt} = \underline{f}(\underline{x}^{(1)}) + \underline{J}(\underline{x}^{(1)}) [\underline{x}^{(2)} - \underline{x}^{(1)}]$$

$$\frac{d\underline{x}^{(2)}}{dt} - \underline{J}(\underline{x}^{(1)}) \underline{x}^{(2)} = \underline{f}(\underline{x}^{(1)}) - \underline{J}(\underline{x}^{(1)}) \underline{x}^{(1)} \quad (137)$$

$$\frac{d\underline{x}^{(2)}}{dt} - \frac{\partial f_1}{\partial x_1} x_1^{(2)} - \frac{\partial f_1}{\partial x_2} x_2^{(2)} - \dots - \frac{\partial f_1}{\partial x_n} x_n^{(2)} = f_1(\underline{x}^{(1)}) - \frac{\partial f_1}{\partial x_1} x_1^{(1)} - \frac{\partial f_1}{\partial x_2} x_2^{(1)} - \dots - \frac{\partial f_1}{\partial x_n} x_n^{(1)}$$

$$\frac{d\underline{x}_i^{(2)}}{dt} = 0, i = 2 \dots n$$

We can not solve this as we don't know the initial condition

We can define a particular solution $\underline{p}(t)$

$$\frac{d\underline{p}}{dt} = \underline{f}(\underline{x}^{(1)}) + \underline{J}^{(1)} [\underline{p} - \underline{x}^{(1)}]$$

$$\underline{p}(0) = \underline{0} \rightarrow \underline{p}_2(t) = \underline{p}_3(t) = \dots \underline{p}_n(t) = 0$$

$$\frac{dP_1}{dt} - \frac{\partial f_1}{\partial x_1} P_1 = f_1^{(1)} - \nabla f_i^T \underline{x}^{(1)}$$

$$P_1(0)=0, \text{ find } P_1(t)=e^{-\frac{\partial f}{\partial x} t} [f_1^{(1)} - \nabla f_i^T \frac{x}{t}]$$

Also, define homogeneous solution

$$\frac{dh^j}{dt} = J^{(1)} \underline{h}^j$$

Obtain $\underline{h}^1(t), \underline{h}^2(t), \dots, \underline{h}^n(t)$.

The solution of eq(137) can be written as

$$\underline{x}^{(2)}(t) = P(t) + \sum_{j=1}^n \underline{h}^j(t) x_j(0) \quad (138)$$

Use (133) to determine

$$\text{Min} \sum_{i=1}^N \left[\hat{\xi}_i - (P_i t_i) + \sum_{j=1}^n h_1^j t_j x_j(0) \right]^2 \quad (139)$$